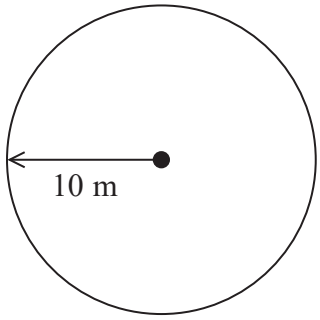
18 Balena has a garden in the shape of a circle of radius 10 m.
He is going to cover the garden with grass seed to make a lawn.

Grass seed is sold in boxes.
Each box of grass seed will cover 46 m^2 of garden.

Balena wants to cover all the garden with grass seed.



(a) Work out an estimate for the number of boxes of grass seed Balena needs.
You must show your working.

Area of circle = $\pi r^2 = \pi \times 10^2$

1 mark
Area formula with r = 10

Estimate $\pi \approx 3$ and $46 \approx 50$.

Estimated area $\approx 3 \times 100 = 300\ m^2$

1 mark
Sensible rounded values

Number of boxes $\approx 300 \div 50 = 6$

1 mark
Divide area by coverage

1 mark
Estimate from 6 to 8 6

(4)

(b) Is your estimate for part (a) an underestimate or an overestimate?
Give a reason for your answer.

An underestimate. Rounding π down makes the area too small. Rounding 46 up makes each box seem to cover more. Both changes reduce the estimate.

1 mark
Reason linked to the estimation choices

(1)

(Total for Question 18 is 5 marks)